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Universitas Sriwijaya

Learning 11

Global State Management

Sharing data across components

Local State vs Shared State

In React, most state lives inside components.

```
function StudentsPage() {
  const [students, setStudents] = useState([])
}
```

This works well for local UI state.

However, some data must be used by many components.

Examples:authentication user, theme (dark / light)

The Props Drilling Problem

When multiple components need the same data, we often pass it via props.

```
<App students={students}>
  <Layout students={students}>
    <Dashboard students={students}>
      <StudentList students={students}/>
```

Solution: Context API

React provides Context API to share data globally.

Context allows components to:

- access shared state
- without passing props manually

Core Concepts of Context API

1. Context

   Creates the shared container

   ```
   const ThemeContext = createContext()
   ```

2. Provider

   Supplies data to the component tree

3. Consumer/useContext

Creating a Context

```
import { createContext } from "react"

export const ThemeContext = createContext()
```

The context acts as a shared data channel.

But by itself, it does not provide data yet.

We need a Provider

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Using a Provider

Provider wraps part of the application

```
<ThemeContext.Provider value={theme}>
  <App />
</ThemeContext.Provider>
```

Now every component inside <App /> can access theme.

Accessing Context with useContext

Components can access shared data using useContext.

```
import { useContext } from "react"

const theme = useContext(ThemeContext)
```

This allows the component to read the global state.

No props required.

Example Use Cases

Common scenarios for Context API:

- theme management
- authentication user
- language settings
- shared application data

Important note: Context API is useful for global or widely shared state.

Challenge! (FE Member Only*)

1. Buat sebuah web (Tema bebas, disarankan melanjutkan project UTS)
2. Implementasikan minimal dua buah Context — satu untuk tema (dark/light mode) dan satu untuk data yang diakses antar komponen
3. Implementasikan apa yang telah dipelajari
4. Lakukan pull request dan masukkan link repo github kamu di folder /11-global-state-management/task.txt
5. Deadline 19 Juni 2026

Selamat Mencoba ~

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Thank You !